Task 1

Dataset description

- The RNA-seq dataset (GSE163190) aims to reveal **how phosphate availability affects chlorophyll accumulation in *Arabidopsis thaliana* (*A. thaliana*) seedlings under iron-deficient conditions** through gene regulation, a biologically important issue as iron and phosphorus are essential materials whose interaction strongly influences **photosynthetic efficiency and nutrient-signaling pathways** in plants.
- The dataset comprises 27 RNA-seq libraries generated from *A. thaliana* grown under three conditions: (+P +Fe), (+P −Fe), and (−P −Fe). For each condition, samples were collected at 39h, 52h, and 76h, with three biological replicates per time point.
- In this project, we focus on the 39h time point, using six libraries in total: **three control samples (PplusFeplus39hours; C1–C3) and three iron-deficient samples (PplusFeminus39hours; T1–T3)**. cDNA libraries were constructed from total RNA and sequenced on an Illumina NextSeq 500 platform in **single-end** mode.

Read Trimming & Quality Control

- Raw reads were trimmed to remove low-quality bases and adapters, and FastQC reports were generated for all six libraries (C1–C3, T1–T3).
- Each library contained **753,088 reads of length 76 bp with a GC content of 44%**. Per-base and per-sequence quality scores were consistently high **(> 30 on average, corresponding to an estimated error rate < 1%)**, and base composition profiles showed only **minor fluctuations at the first few cycles before stabilizing at ~20–30% for each nucleotide**.
- Overall, the six RNA-seq libraries are of high quality and suitable for downstream analyses. Although some metrics are denoted as not perfect by the strict criteria of FastQC (e.g., sequence duplication of poly-A), we still consider these sequence files to be quite acceptable data.

Read Alignment & Mapping Statistics

- RNA-seq reads from the six libraries were aligned to the TAIR10 reference genome (release 52) using the corresponding FASTA sequence and GTF gene annotation.
- Across all samples, alignment quality was high, with **more than 90% of reads uniquely mapped** in each library, indicating reliable sequencing and an appropriate choice of reference. Approximately **7% of reads mapped to multiple loci**, which is expected given the prevalence of homologous genes and gene families in plant genomes. Around **1% of reads remained unmapped**, mainly due to short or low-complexity sequences (Table 1).
- These mapping statistics indicate that the alignments are of high quality, with minimal technical noise, and are suitable for downstream transcriptomic analyses.

Samples	*Uniquely mapped (%)*	*Multi-mapped (%)*	*Unmapped (%)*
C1-C3	***90.95***	***7.89***	***1.16***
T1-T3	***91.10***	***7.63***	***1.33***

Table 1. *Read alignment summary for the six RNA-seq libraries. Average values are presented.*

Task 2

Reference-based Transcriptome Reconstruction

- By selecting C1 and performing reference-guided transcriptome assembly using StringTie, we obtained a reassembled transcriptome annotation in GTF format, along with a gene-level abundance table.

Reconstructed transcriptome and comparison with known annotations

- Through some statistical analyses, we found that StringTie assembly for sample **C1** yielded **15,108 transcripts comprising 49,779 exons**. At the gene level, **32,832 known genes** are reported.
- These basic statistics suggest that the reference-guided assembly is reliable and does not introduce obvious artefactual transcript structures at the inspected loci.
- In IGV, we compared the reconstructed transcriptome to a reference transcriptome and selected three representative types of loci.

    - Scanning genes **with high TPM and a single dominant isoform (robust genes)**, we detected a gene that closely matches the reference annotation (Figure 1).
    - Scanning genes **with multiple assembled isoforms (sections vary extensively)**, we detected loci where alternative splicing or structural divergence occur (Figure 2).
    - Scanning genes **without known gene-IDs with relatively high expression (robust genes disappearing in the known transcriptomes)**, we detected transcripts without corresponding entries in the reference annotation. (Figure 3).

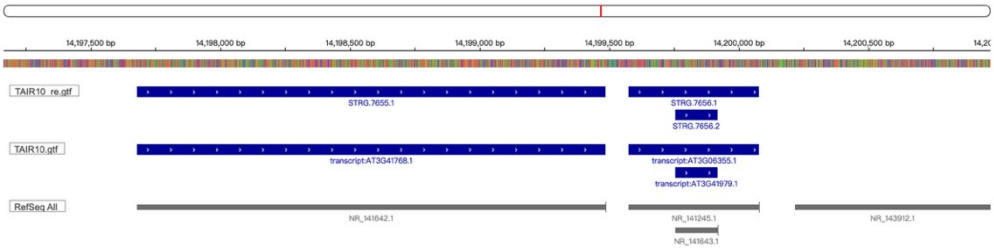


Figure 1. ***AT3G41768****). The gene exhibits the expected exon–intron structure, as a reference.*

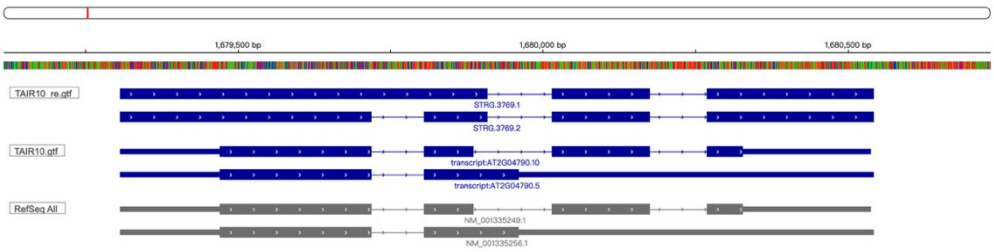


Figure 2. ***AT2G04790****. The reconstructed transcripts reveal shifts in exon boundaries compared to the reference.*

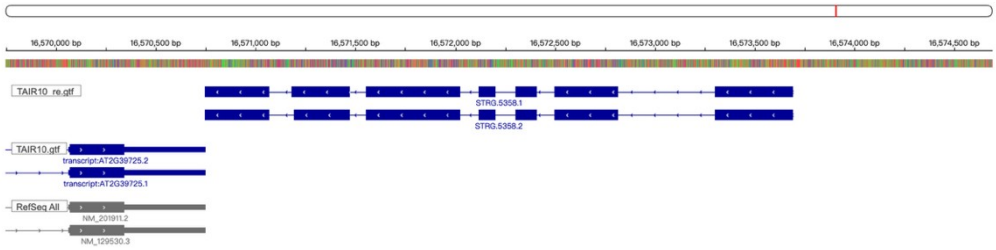


Figure 3. ***STRG.5358****. Novel transcripts were reported only in the reconstructed transcriptome at track 1.*

Task 3

DEG Analysis

- Counts from featureCounts were imported into R and analyzed with DESeq2.
- Using a threshold of adjusted p-value < 0.05 and $|\log_2$ fold change$|$ > 1, we detected **161 up-regulated and 62 down-regulated genes**. MA and volcano plots (Figure 4 & 5) displayed the DEGs and their distributions in counts/FDR levels.

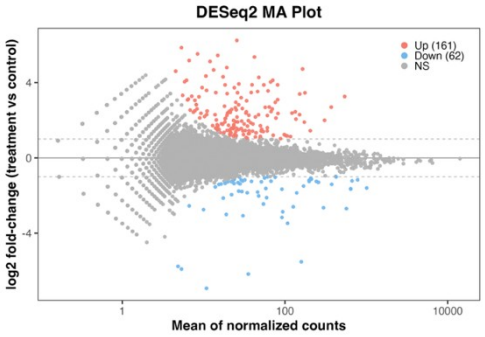


Figure 4. Mean–average (MA) plot.

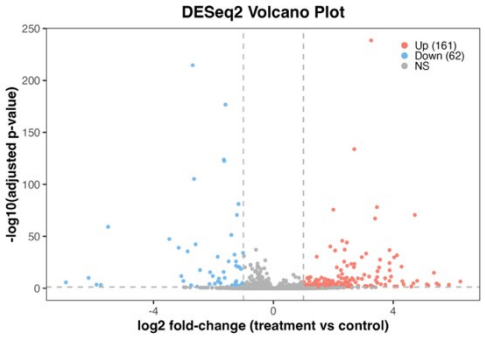


Figure 5. Volcano plot.

IGV Verification

- IGV inspection was consistent with the DEG. With all tracks normalized to the same coverage range, AT5G53450 (Figure 6) showed higher read expression in the treatment group than in controls, whereas AT4G25100 (Figure 7) showed a reduced pattern.

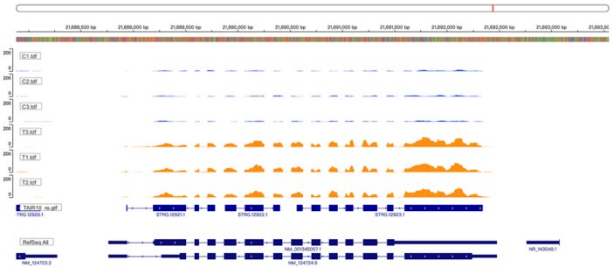


Figure 6. AT5G53450 (up-regulated). Control (C1–C3, blue) and treatment (T1–T3, orange). All normalized to the same scale.

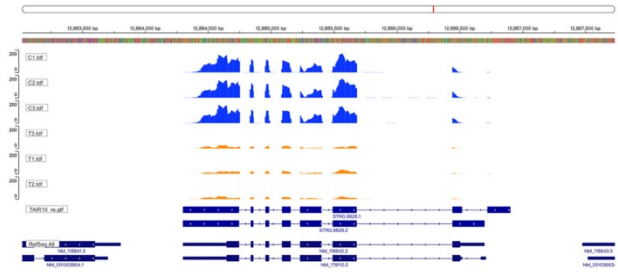


Figure 7. AT4G25100 (down-regulated). Control (C1–C3, blue) and treatment (T1–T3, orange). All normalized to the same scale.

Functional Enrichment ***(up- & down-regulated genes are combined for simplicity)***

- Using clusterProfiler, we identified **68 significantly enriched GO terms (59 BP, 5 CC, 4 MF)** (Figure 8) and **8 KEGG pathways** (Figure 9).

- Enriched GO terms include ***response to hypoxia (BP), photosystem I (CC), chlorophyll binding (MF)***, and KEGG pathways involve **photosynthesis.**
- These consistently highlight **photosynthesis-related functions and pathways**, indicating that *A. thaliana* integrates phosphorus and iron availability by modulating genes involved in photosynthesis under nutrient deficiency.

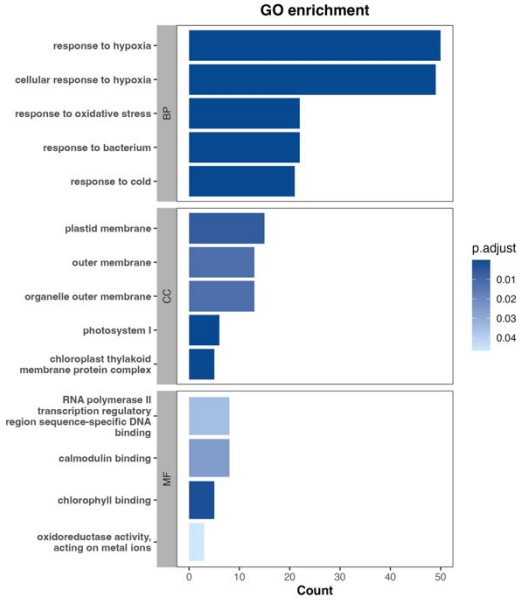


Figure 8. GO enrichment analysis for DEGs,
(FDR < 0.05 and |LFC| ≥ 1), showing the top terms for Biological Process (BP), Cellular Component (CC) and Molecular Function (MF). Bars indicate the number of DEGs, and colors encode the adjusted p-value.

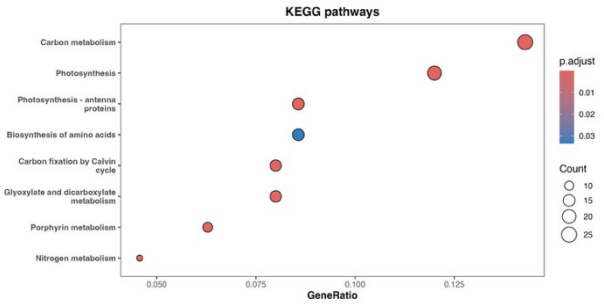


Figure 9. KEGG pathway analysis for DEGs. K (FDR < 0.05 and |LFC| ≥ 1). Points' positions indicate the gene ration, colors encode the adjusted p-value, and dot sizes illustrate the number of DEGs.

Task 4

Introduction

- RNA sequencing (RNA-seq) is an experimental–computational workflow that converts cellular RNA into sequenceable molecules and quantifies transcript abundance.
- The core stage of RNA-Seq involves next-generation sequencing (NGS), which utilizes Illumina platforms widely. Further data analyses can help interpret the results.

Relation to gene regulation

- RNA-seq quantifies differential gene expression across conditions, allowing for the inference of gene regulatory mechanisms through multiple post-analysis methods.
- Here are some brief examples: **Normalized expression measures and DEG analysis** enable the valid comparison of gene expression levels among samples, which can reveal gene regulation at the transcript abundance level in a statistically significant manner. In addition, **functional enrichment** further clusters the DEGs into terms or pathways related to **specific biological meanings**, revealing or verifying the genes and regulations at a more detailed level. Moreover, **alternative splicing analysis** detects splicing exons, junctions, or isomers, hence quantifying gene expression at the transcript level. These help reveal **post-transcriptional gene regulation**.

Essential Steps – Well-developed Strategies & Remaining Limitations

- For sample preparation, **careful experimental design** is crucial, such as using multiple biological replicates. For modern Illumina RNA-seq, **technical variability** is slight, so resources are better allocated to biological replicates. **RNA selection**, **fragmentation, and priming strategies**, **as well as pairing strategies and strand-specific protocols**, must match the quality of the RNA and the biological questions being addressed.

- Computationally, **quality control and trimming** are now routine. Reads are aligned to **increasingly refined reference genomes** and **quantified with mature tools** with **unified measure indices**. **Information extraction pipelines**, including DEG, functional enrichment, alternative splicing analysis, and visualization methods, are well-supported by software. However, practical challenges such as short-read ambiguity still exist.

Future Challenges & Promising Directions

- **Direct RNA sequencing** (e.g., using Oxford Nanopore) profiles RNA molecules without requiring reverse transcription, thereby preserving more native signals, such as **base modifications** (e.g., m^6A). When these raw signals are **combined with machine-learning models**, they can be used to **predict sequences and modification sites with greater precision**. Joint analysis with epigenomic assays also helps **relate post-transcriptional regulations to chromatin and regulatory context**.
- **Improvements in statistical interpretations are urgently needed.** When interpreting analysis results, **greater weight should be placed on reproducibility across replicates and on the biological interpretability of the results, rather than relying solely on single significance thresholds (e.g., p-value = 0.05)**.
- **Spatial transcriptomic assays** retain information on tissue architecture and spatial coordinates, allowing an integrated **"gene–cell–environment" working framework**.

Appendix (all codes used are listed here)

- **Linux**

Step 1 - Download FASTQ files

```
1. wget -c http://10.7.88.74/MyWeb/RNAseq/C1.fastq
2. wget -c http://10.7.88.74/MyWeb/RNAseq/C2.fastq
3. wget -c http://10.7.88.74/MyWeb/RNAseq/C3.fastq
4. wget -c http://10.7.88.74/MyWeb/RNAseq/T1.fastq
5. wget -c http://10.7.88.74/MyWeb/RNAseq/T2.fastq
6. wget -c http://10.7.88.74/MyWeb/RNAseq/T3.fastq
7. wget -c http://10.7.88.74/MyWeb/RNAseq/TAIR10.gtf
8. wget -c http://10.7.88.74/MyWeb/RNAseq/TAIR10.chrom.sizes
9. wget -c http://10.7.88.74/MyWeb/RNAseq/TAIR10.toplevel.fa
```

Step 2 - Quality Control

```
1. fastqc -o ~/bio211/W7/Part2/step2/fastqc_report \
```

```
C1.fastq C2.fastq \
C3.fastq T1.fastq \
T2.fastq T3.fastq
```

Step 3 - Trim

```
trim_galore C1.fastq -o ~/bio211/W7/Part2/step3/trimed/ &
trim_galore C2.fastq -o ~/bio211/W7/Part2/step3/trimed/ &
trim_galore C3.fastq -o ~/bio211/W7/Part2/step3/trimed/ &
trim_galore T1.fastq -o ~/bio211/W7/Part2/step3/trimed/ &
trim_galore T2.fastq -o ~/bio211/W7/Part2/step3/trimed/ &
trim_galore T3.fastq -o ~/bio211/W7/Part2/step3/trimed/ &
```

Step 4 - Make Index

```
STAR --runMode genomeGenerate \
--genomeDir ~/bio211/W7/Part2/step4/index/ --runThreadN 10 \
--genomeFastaFiles ~/bio211/W7/Part2/step1/TAIR10.toplevel.fa \
--sjdbGTFfile ~/bio211/W7/Part2/step1/TAIR10.gtf \
--genomeSAindexNbases 12
```

Step 5 - Alignment

```
STAR --outSAMtype BAM SortedByCoordinate --runThreadN 5 \
    --genomeDir ~/bio211/W7/Part2/step4/index/ --outFileNamePrefix
    ~/bio211/W7/Part2/step5/bam/C1 \
    --readFilesIn ~/bio211/W7/Part2/step3/trimed/C1_trimmed.fq &
STAR --outSAMtype BAM SortedByCoordinate --runThreadN 5 \
```

```
    --genomeDir ~/bio211/W7/Part2/step4/index/ --outFileNamePrefix
    ~/bio211/W7/Part2/step5/bam/C2 \
    --readFilesIn ~/bio211/W7/Part2/step3/trimed/C2_trimmed.fq &
STAR --outSAMtype BAM SortedByCoordinate --runThreadN 5 \
    --genomeDir ~/bio211/W7/Part2/step4/index/ --outFileNamePrefix
    ~/bio211/W7/Part2/step5/bam/C3 \
    --readFilesIn ~/bio211/W7/Part2/step3/trimed/C3_trimmed.fq &
STAR --outSAMtype BAM SortedByCoordinate --runThreadN 5 \
    --genomeDir ~/bio211/W7/Part2/step4/index/ --outFileNamePrefix
    ~/bio211/W7/Part2/step5/bam/T1 \
    --readFilesIn ~/bio211/W7/Part2/step3/trimed/T1_trimmed.fq &
STAR --outSAMtype BAM SortedByCoordinate --runThreadN 5 \
    --genomeDir ~/bio211/W7/Part2/step4/index/ --outFileNamePrefix
    ~/bio211/W7/Part2/step5/bam/T2 \
    --readFilesIn ~/bio211/W7/Part2/step3/trimed/T2_trimmed.fq &
STAR --outSAMtype BAM SortedByCoordinate --runThreadN 5 \
    --genomeDir ~/bio211/W7/Part2/step4/index/ --outFileNamePrefix
    ~/bio211/W7/Part2/step5/bam/T3 \
    --readFilesIn ~/bio211/W7/Part2/step3/trimed/T3_trimmed.fq &
```

Step 6 - Visualization Preparation

```
igvtools count -z 5 -w 10 -e 0
    ~/bio211/W7/Part2/step5/bam/C1Aligned.sortedByCoord.out.bam
```

```
    ~/bio211/W7/Part2/step6/tdf/C1.tdf
    ~/bio211/W7/Part2/step1/TAIR10.chrom.sizes &
igvtools count -z 5 -w 10 -e 0
    ~/bio211/W7/Part2/step5/bam/C2Aligned.sortedByCoord.out.bam
    ~/bio211/W7/Part2/step6/tdf/C2.tdf
    ~/bio211/W7/Part2/step1/TAIR10.chrom.sizes &
igvtools count -z 5 -w 10 -e 0
    ~/bio211/W7/Part2/step5/bam/C3Aligned.sortedByCoord.out.bam
    ~/bio211/W7/Part2/step6/tdf/C3.tdf
    ~/bio211/W7/Part2/step1/TAIR10.chrom.sizes &
igvtools count -z 5 -w 10 -e 0
    ~/bio211/W7/Part2/step5/bam/T1Aligned.sortedByCoord.out.bam
    ~/bio211/W7/Part2/step6/tdf/T1.tdf
    ~/bio211/W7/Part2/step1/TAIR10.chrom.sizes &
igvtools count -z 5 -w 10 -e 0
    ~/bio211/W7/Part2/step5/bam/T2Aligned.sortedByCoord.out.bam
    ~/bio211/W7/Part2/step6/tdf/T2.tdf
    ~/bio211/W7/Part2/step1/TAIR10.chrom.sizes &
igvtools count -z 5 -w 10 -e 0
    ~/bio211/W7/Part2/step5/bam/T3Aligned.sortedByCoord.out.bam
    ~/bio211/W7/Part2/step6/tdf/T3.tdf
    ~/bio211/W7/Part2/step1/TAIR10.chrom.sizes &
```

Step7 - Reference based transcriptome reconstruction

```
stringtie
    ~/bio211/W7/Part2/step5/bam/C1Aligned.sortedByCoord.out.bam -G
    ~/bio211/W7/Part2/step1/TAIR10.gtf
    -o ~/bio211/W7/Part2/step7/gtf/TAIR10_re.gtf
    -A ~/bio211/W7/Part2/step7/gtf/C1.txt -p 8
```

Step8 - Gene expression estimate by featureCounts

```
featureCounts -t exon -g gene_id \
    -a ~/bio211/W7/Part2/step1/TAIR10.gtf
    -o ~/bio211/W7/Part2/step8/exp/feature.txt \
    ~/bio211/W7/Part2/step5/bam/C1Aligned.sortedByCoord.out.bam \
    ~/bio211/W7/Part2/step5/bam/C2Aligned.sortedByCoord.out.bam \
    ~/bio211/W7/Part2/step5/bam/C3Aligned.sortedByCoord.out.bam \
    ~/bio211/W7/Part2/step5/bam/T1Aligned.sortedByCoord.out.bam \
    ~/bio211/W7/Part2/step5/bam/T2Aligned.sortedByCoord.out.bam \
    ~/bio211/W7/Part2/step5/bam/T3Aligned.sortedByCoord.out.bam
```

- **R**

Step 9 – DEG analysis

```
library(DESeq2)
library(ggplot2)
library(grid)

# --- Read featureCounts output and prepare count matrix ---
```

```
6.
7.  sampleNames <- c("C1", "C2", "C3", "T1", "T2", "T3")
8.  ED1 <- read.table("feature.txt", header = TRUE, quote = "\t",
9.  skip = 1) # Expression Data 1
10.
11.  names(ED1)[c(7:12)] <- sampleNames
12.  countData <- as.matrix(ED1[7:12])
13.  rownames(countData) <- ED1$Geneid
14.
15.  # Remove genes with zero counts across all samples
16.  countData <- countData[rowSums(countData) != 0, ]
17.
18.  # --- Sample metadata ---
19.
20.  SRR_df <- data.frame(
21.    id              = c("C1", "C2", "C3", "T1", "T2", "T3"),
22.    condition = c("control", "control", "control",
23.                  "treatment", "treatment", "treatment")
24.  )
25.
26.  # Check that column names match sample IDs
27.  all(colnames(countData) == SRR_df$id)
```

```
28.
29.  # --- DESeq2 differential expression analysis ---
30.
31.  dds <- DESeqDataSetFromMatrix(
32.      countData = countData,
33.      colData   = SRR_df,
34.      design    = ~ condition
35.  )
36.  dds <- DESeq(dds)
37.  res <- results(dds)    # treatment / control
38.
39.  res
40.  sum(res$padj < 0.05, na.rm = TRUE)
41.
42.  # --- MA plot (FDR < 0.05 & |LFC| ≥ 1 highlighted) ---
43.
44.  ma_df   <- as.data.frame(res)
45.  alpha   <- 0.05
46.  lfc_cut <- 1
47.
48.  ma_df$dir <- with(
49.      ma_df,
```

```
    ifelse(!is.na(padj) & padj < alpha & log2FoldChange >=
lfc_cut, "Up",
    ifelse(!is.na(padj) & padj < alpha & log2FoldChange <= -
lfc_cut, "Down", "NS"))
)
ma_df$dir <- factor(ma_df$dir, levels = c("Up", "Down", "NS"))

n_up   <- sum(ma_df$dir == "Up",   na.rm = TRUE)
n_down <- sum(ma_df$dir == "Down", na.rm = TRUE)

p_ma <- ggplot(ma_df, aes(x = baseMean, y = log2FoldChange,
color = dir)) +
  geom_hline(
    yintercept = c(-1, 0, 1),
    linetype   = c(2, 1, 2),
    linewidth  = c(0.3, 0.4, 0.3),
    color      = c("grey70", "grey60", "grey70")
  ) +
  geom_point(alpha = 0.9, size = 0.85, na.rm = TRUE) +
  scale_x_log10() +
  scale_color_manual(
    values = c(Up = "#f57c6e", Down = "#71b7ed", NS = "grey70"),
```

```
    breaks = c("Up", "Down", "NS"),
    labels = c(
      paste0("Up (",      n_up,      ")"),
      paste0("Down (", n_down, ")"),
      "NS"
    ),
    name = NULL
  ) +
  labs(
    x        = "Mean of normalized counts",
    y        = "log2 fold-change (treatment vs control)",
    title = "DESeq2 MA Plot"
  ) +
  theme_bw(base_size = 12) +
  theme(
    panel.grid              = element_blank(),
    legend.position      = c(0.98, 0.98),
    legend.justification = c(1, 1),
    plot.title              = element_text(hjust = 0.5, face = "bold"),
    axis.title.x          = element_text(face = "bold"),
    axis.title.y          = element_text(face = "bold")
  ) +
```

```
    guides(
      color = guide_legend(
        title = NULL,
        override.aes = list(size = 2, alpha = 1)
      )
    ) +
    theme(
      legend.key.height = grid::unit(6, "pt"),
      legend.key.width  = grid::unit(6, "pt")
    )

print(p_ma)
ggsave("MAplot.png", p_ma, width = 6, height = 4.2, dpi = 300)

# --- Merge DESeq2 results with normalized counts ---

resdata <- merge(
  as.data.frame(res),
  as.data.frame(counts(dds, normalized = TRUE)),
  by    = "row.names",
  sort = FALSE
)
```

```
colnames(resdata)[1] <- "gene_id"

# Tag genes as Up / Down / NoDiff
resdata$change <- as.factor(
    ifelse(resdata$padj < 0.05 & abs(resdata$log2FoldChange) > 1,
                ifelse(resdata$log2FoldChange > 1, "Up", "Down"),
                "NoDiff")
)

upRegu      <- resdata[which(resdata$change == "Up"), ]
downRegu <- resdata[which(resdata$change == "Down"), ]

print(paste0("Up-regulated genes: ",      nrow(upRegu)))
print(paste0("Down-regulated genes: ", nrow(downRegu)))

# Remove "gene:" prefix for output
downRegu$gene_id <- sub("gene:", "", downRegu$gene_id)
upRegu$gene_id      <- sub("gene:", "", upRegu$gene_id)

# write.table(upRegu$gene_id,
#                     "upRegulateGene_0.05.txt",
#                     col.names = FALSE,
```

```
#                   row.names = FALSE,
#                   quote     = FALSE)
# write.table(downRegu$gene_id,
#                   "downRegulateGene_0.05.txt",
#                   col.names = FALSE,
#                   row.names = FALSE,
#                   quote     = FALSE)

# --- Volcano plot ---

resdata$change <- factor(resdata$change,
                         levels = c("Up", "Down", "NoDiff"))

alpha_thr <- 0.05

volcano <- ggplot(
  resdata,
  aes(x = log2FoldChange,
      y = -log10(padj),
      color = change)
) +
  geom_point(alpha = 0.9, size = 0.9, na.rm = TRUE) +
```

```
    geom_vline(
      xintercept = c(-1, 1),
      linetype   = 2,
      color      = "grey70"
    ) +
    geom_hline(
      yintercept = -log10(alpha_thr),
      linetype   = 2,
      color      = "grey70"
    ) +
    scale_color_manual(
      values = c(
        Up     = "#f57c6e",
        Down   = "#71b7ed",
        NoDiff = "grey70"
      ),
      breaks = c("Up", "Down", "NoDiff"),
      labels = c(
        paste0("Up (",   n_up,   ")"),
        paste0("Down (", n_down, ")"),
        "NS"
      ),
```

```
    name = NULL
  ) +
  labs(
    x        = "log2 fold-change (treatment vs control)",
    y        = "-log10(adjusted p-value)",
    title = "DESeq2 Volcano Plot"
  ) +
  theme_bw(base_size = 12) +
  theme(
    panel.grid          = element_blank(),
    legend.position     = c(0.98, 0.98),
    legend.justification = c(1, 1),
    plot.title          = element_text(hjust = 0.5, face = "bold"),
    axis.title.x        = element_text(face = "bold"),
    axis.title.y        = element_text(face = "bold"),
    legend.key.height = grid::unit(6, "pt"),
    legend.key.width  = grid::unit(6, "pt")
  ) +
  guides(
    color = guide_legend(
      title = NULL,
      override.aes = list(size = 2, alpha = 1)
```

```
204.          )
205.      )
206.
207.  print(volcano)
208.  ggsave("Volcano.png", volcano, width = 6, height = 4.2, dpi =
209.  300)
```

Step 10 – GO/KEGG analysis

```
1.  library(DESeq2)
2.  library(ggplot2)
3.  library(clusterProfiler)
4.  library(enrichplot)
5.  library(org.At.tair.db)
6.  library(dplyr)
7.  library(stringr)
8.
9.  # --- Read featureCounts output and run DESeq2 (as in Step 10)
10. ---
11.
12. sampleNames <- c("C1", "C2", "C3", "T1", "T2", "T3")
13. ED1 <- read.table("./feature.txt", header = TRUE, quote = "\t",
14. skip = 1)
15.
```

```
16.  names(ED1)[c(7:12)] <- sampleNames
17.  countData <- as.matrix(ED1[7:12])
18.  rownames(countData) <- ED1$Geneid
19.
20.  # Remove genes with zero total counts
21.  countData <- countData[rowSums(countData) != 0, ]
22.
23.  SRR_df <- data.frame(
24.    id              = c("C1", "C2", "C3", "T1", "T2", "T3"),
25.    condition = c("control", "control", "control",
26.                          "treatment", "treatment", "treatment")
27.  )
28.
29.  all(colnames(countData) == SRR_df$id)
30.
31.  dds <- DESeqDataSetFromMatrix(
32.    countData = countData,
33.    colData     = SRR_df,
34.    design       = ~ condition
35.  )
36.  dds <- DESeq(dds)
37.  res <- results(dds)    # treatment / control
```

```
38.
39.  # --- Merge DESeq2 results with normalized counts ---
40.
41.  resdata <- merge(
42.      as.data.frame(res),
43.      as.data.frame(counts(dds, normalized = TRUE)),
44.      by      = "row.names",
45.      sort = FALSE
46.  )
47.  colnames(resdata)[1] <- "gene_id"
48.
49.  # Tag genes as Up / Down / NoDiff (for reference)
50.  resdata$change <- as.factor(
51.      ifelse(resdata$padj < 0.05 & abs(resdata$log2FoldChange) > 1,
52.                  ifelse(resdata$log2FoldChange > 1, "Up", "Down"),
53.                  "NoDiff")
54.  )
55.  resdata$change <- factor(resdata$change,
56.                                      levels = c("Up", "Down", "NoDiff"))
57.
58.  # --- Helper functions for faceted GO barplot ---
59.
```

```
reorder_within <- function(x, by, within, fun = mean, sep = "___", ...) {
  new_x <- paste(x, within, sep = sep)
  stats::reorder(new_x, by, FUN = fun)
}

scale_y_reordered <- function(..., sep = "___") {
  ggplot2::scale_y_discrete(
    labels = function(x) gsub(paste0(sep, ".+$"), "", x),
    ...
  )
}

# --- Define DEG sets and background ---

all_ids <- sub("^gene:", "", resdata$gene_id)

deg_flag <- !is.na(resdata$padj) &
                resdata$padj < 0.05 &
                abs(resdata$log2FoldChange) >= 1

up_ids      <- sub("^gene:", "", resdata$gene_id[deg_flag &
resdata$log2FoldChange >    0])
```

```
down_ids <- sub("^gene:", "", resdata$gene_id[deg_flag &
resdata$log2FoldChange <    0])
deg_all    <- unique(c(up_ids, down_ids))

universe_ids <- unique(all_ids)

# --- GO enrichment (BP / CC / MF) ---

ego_bp <- enrichGO(
    gene                 = deg_all,
    universe          = universe_ids,
    OrgDb               = org.At.tair.db,
    keyType            = "TAIR",
    ont                   = "BP",
    pAdjustMethod = "BH",
    pvalueCutoff    = 0.05,
    qvalueCutoff    = 0.05
)

ego_cc <- enrichGO(
    gene                 = deg_all,
    universe          = universe_ids,
```

```
    OrgDb         = org.At.tair.db,
    keyType       = "TAIR",
    ont           = "CC",
    pAdjustMethod = "BH",
    pvalueCutoff  = 0.05,
    qvalueCutoff  = 0.05
)

ego_mf <- enrichGO(
    gene          = deg_all,
    universe      = universe_ids,
    OrgDb         = org.At.tair.db,
    keyType       = "TAIR",
    ont           = "MF",
    pAdjustMethod = "BH",
    pvalueCutoff  = 0.05,
    qvalueCutoff  = 0.05
)

# Remove redundant GO terms
ego_bp <- tryCatch(
    simplify(ego_bp, cutoff = 0.7, by = "p.adjust", select_fun =
```

```
min),
    error = function(e) ego_bp
)
ego_cc <- tryCatch(
    simplify(ego_cc, cutoff = 0.7, by = "p.adjust", select_fun =
min),
    error = function(e) ego_cc
)
ego_mf <- tryCatch(
    simplify(ego_mf, cutoff = 0.7, by = "p.adjust", select_fun =
min),
    error = function(e) ego_mf
)

# --- Prepare GO results for plotting ---

df_bp <- transform(as.data.frame(ego_bp), ONTO = "BP")
df_cc <- transform(as.data.frame(ego_cc), ONTO = "CC")
df_mf <- transform(as.data.frame(ego_mf), ONTO = "MF")

g_all <- dplyr::bind_rows(df_bp, df_cc, df_mf)

```

```
g_all$GeneRatio_num <- if (is.numeric(g_all$GeneRatio)) {
  g_all$GeneRatio
} else {
  vapply(
    strsplit(as.character(g_all$GeneRatio), "/"),
    function(x) as.numeric(x[1]) / as.numeric(x[2]),
    numeric(1)
  )
}

g_bp_top <- g_all[g_all$ONTO == "BP", ] |>
  dplyr::arrange(dplyr::desc(GeneRatio_num)) |>
  dplyr::slice_head(n = 5)

g_cc_top <- g_all[g_all$ONTO == "CC", ] |>
  dplyr::arrange(dplyr::desc(GeneRatio_num)) |>
  dplyr::slice_head(n = 5)

g_mf_top <- g_all[g_all$ONTO == "MF", ] |>
  dplyr::arrange(dplyr::desc(GeneRatio_num)) |>
  dplyr::slice_head(n = 5)

```

```
g_top <- dplyr::bind_rows(g_bp_top, g_cc_top, g_mf_top)

g_top$Description <- stringr::str_wrap(g_top$Description, width
= 28)

p_go_bar <- ggplot(
    g_top,
    aes(x = Count,
        y = reorder_within(Description, GeneRatio_num, ONTO),
        fill = p.adjust)
) +
    geom_col() +
    scale_y_reordered() +
    facet_grid(ONTO ~ ., scales = "free_y", switch = "y") +
    scale_fill_gradient(
        name = "p.adjust",
        low  = "#cfe8ff",
        high = "#084a91",
        trans = "reverse"
    ) +
    labs(
        x        = "Count",
```

```
        y         = NULL,
        title = "GO enrichment"
      ) +
      theme_bw(base_size = 12) +
      theme(
        panel.grid        = element_blank(),
        strip.background = element_rect(fill = "grey70", color = NA),
        plot.title        = element_text(hjust = 0.5, face = "bold"),
        axis.title.x    = element_text(face = "bold"),
        axis.title.y    = element_text(face = "bold"),
        axis.text.y      = element_text(face = "bold")
      )

p_go_bar
ggsave("GO.png", p_go_bar, width = 7, height = 8, dpi = 300)

# --- KEGG enrichment (organism = 'ath'; ENTREZ IDs required) -
--

deg_all <- sub("^gene:", "",
resdata$gene_id[!is.na(resdata$padj) & resdata$padj < 0.05])

```

```
map_all <- bitr(
    deg_all,
    fromType = "TAIR",
    toType      = "ENTREZID",
    OrgDb       = org.At.tair.db
)
entrez <- unique(na.omit(map_all$ENTREZID))

ekegg <- enrichKEGG(
    gene                = entrez,
    organism            = "ath",
    keyType             = "ncbi-geneid",
    pvalueCutoff    = 0.05,
    pAdjustMethod = "BH"
)

n <- nrow(as.data.frame(ekegg))
n

p_kegg <- enrichplot::dotplot(ekegg, showCategory = n) +
    scale_color_continuous(name = "p.adjust", guide = "legend") +
    labs(
```

```
        x         = "GeneRatio",
        y         = NULL,
        title = "KEGG pathways"
    ) +
    theme_bw(base_size = 12) +
    theme(
        panel.grid      = element_blank(),
        plot.title      = element_text(hjust = 0.5, face = "bold"),
        axis.title.x = element_text(face = "bold"),
        axis.title.y = element_text(face = "bold"),
        axis.text.y    = element_text(face = "bold")
    )

p_kegg
ggsave("KEGG.png", p_kegg, width = 10, height = 5, dpi = 300)

# --- Console summary of enrichment counts ---

cat(sprintf("GO terms: BP=%d, CC=%d, MF=%d; KEGG=%s\n",
                    nrow(as.data.frame(ego_bp)),
                    nrow(as.data.frame(ego_cc)),
                    nrow(as.data.frame(ego_mf)),
```

```
258.                         if (is.null(ekegg)) "NA" else
259.   as.character(nrow(as.data.frame(ekegg)))))
```